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Control Study of a Differential Drive Mobile Platform

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Academic Year: 2024 – 2025

Introduction

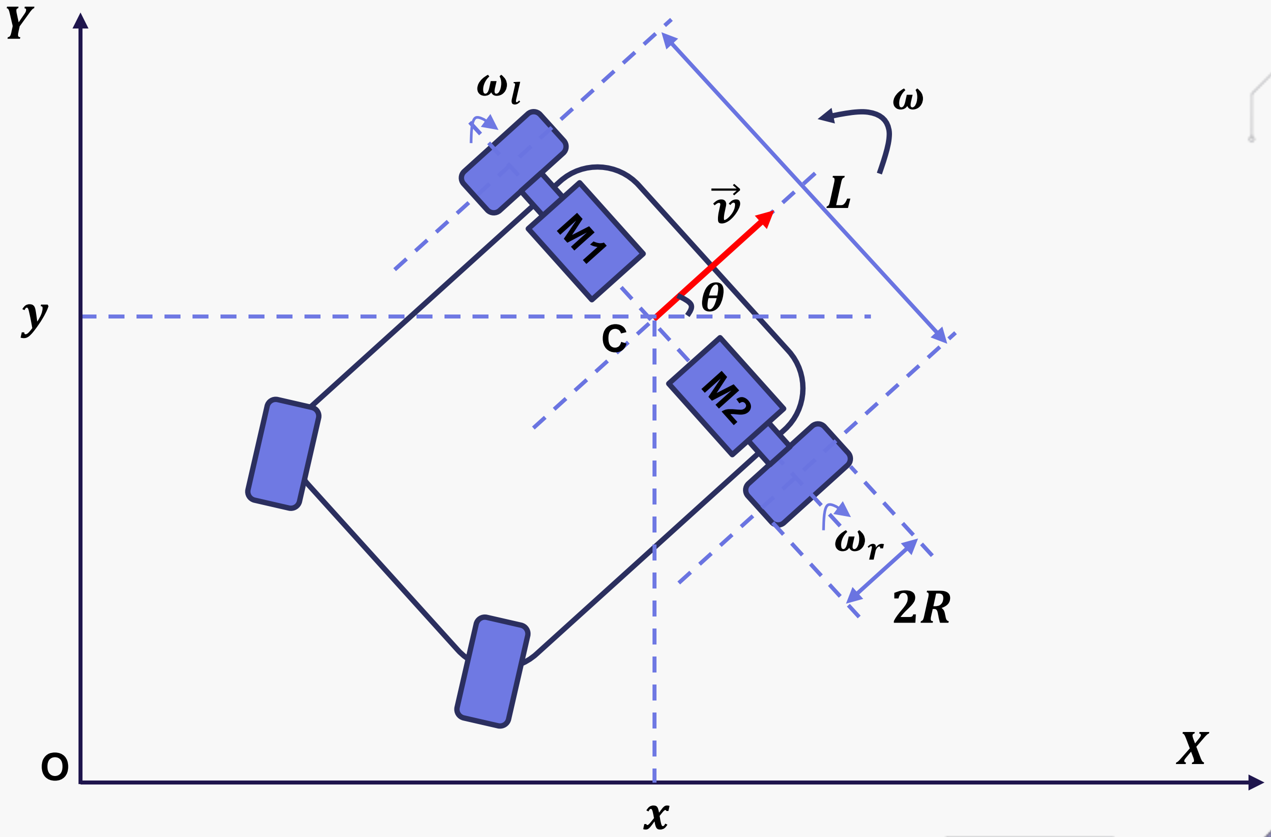
Differential-drive mobile robots are widely used due to their simple structure and high maneuverability.

These robots use two independently powered wheels to control movement and direction without mechanical steering systems.

This project aims to digitally replicate the function of a mechanical differential by controlling each motor individually.

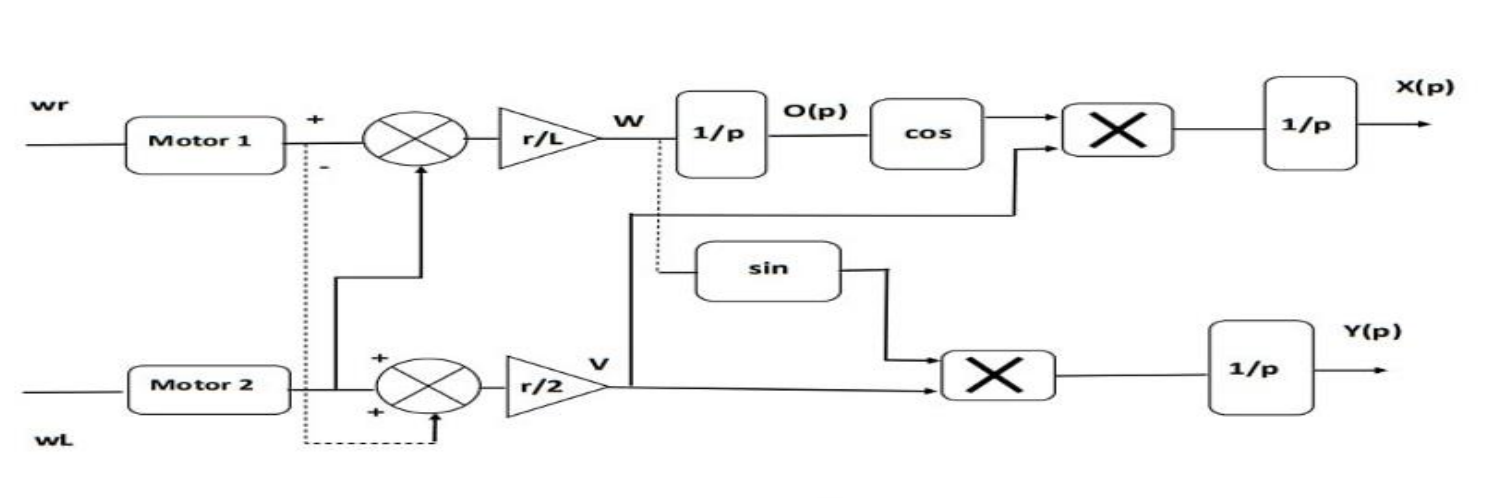
The goal is to study by simulation control strategies for the robot's movement using MATLAB/Simulink.

Motion Model of the Differential Drive Mobile Robot (DDMR)



Motion Model of the Differential Drive Mobile Robot (DDMR)

This diagram shows the simulation model of the robot, combining the motion equations, DC motor dynamics, and PID controllers.



It allows testing the response of the full system to different speed and trajectory inputs.

Simple Motion Control Strategy – Open-Loop

In a simple open-loop strategy, the robot's movement is controlled by directly setting the wheel speeds without using any feedback.
Different combinations of left and right wheel speeds (ω_l and ω_r) produce different types of motion:

- ⬆️ $\omega r = \omega l$ → Straight-line motion (forward)
- ↩️ $\omega r > \omega l$ → Curved trajectory turning left
- ↪️ $\omega l > \omega r$ → Curved trajectory turning right
- 🔄 $\omega r = -\omega l$ → In-place rotation (turns on the spot)

Mechanical and Electrical Modeling of DC Motors

To understand the dynamics of the mobile robot, we start by modeling the motors that drive its wheels using its electrical and mechanichal equations

Electrical Eq: $U(t) = Ri(t) + L\frac{di(t)}{dt} + K_e\omega(t)$

Mechanical Eq: $J\frac{d\omega(t)}{dt} + f\omega(t) = C_m(t) - C_r(t)$

These differential equations are then converted into the Laplace domain. This transformation simplifies the analysis and allows for block diagram representation and control design.

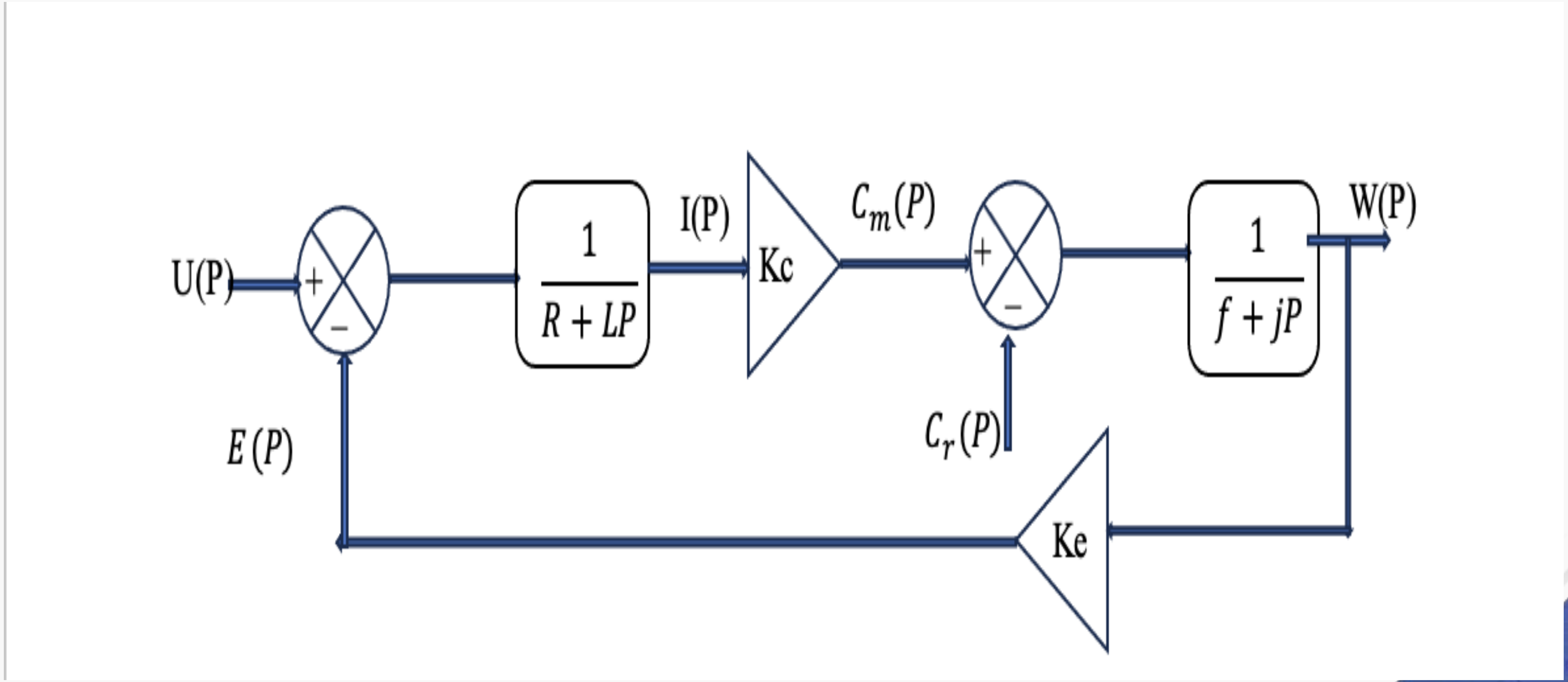
Laplace Domain:

$$C_m(p) - C_r(P) = (Jp + f)\omega(p)$$

$$U(P) = (\mathrm{R} + \mathrm{Lp})\mathrm{i(p)} + \mathrm{K_e}\omega(p)$$

Mechanical and Electrical Modeling of DC Motors

Functional block diagram of DC motor.



PID Speed Regulation of DC Motors

To regulate the angular speed of the DC motors, we used a PID controller based on speed error

Error = desired speed - measured speed

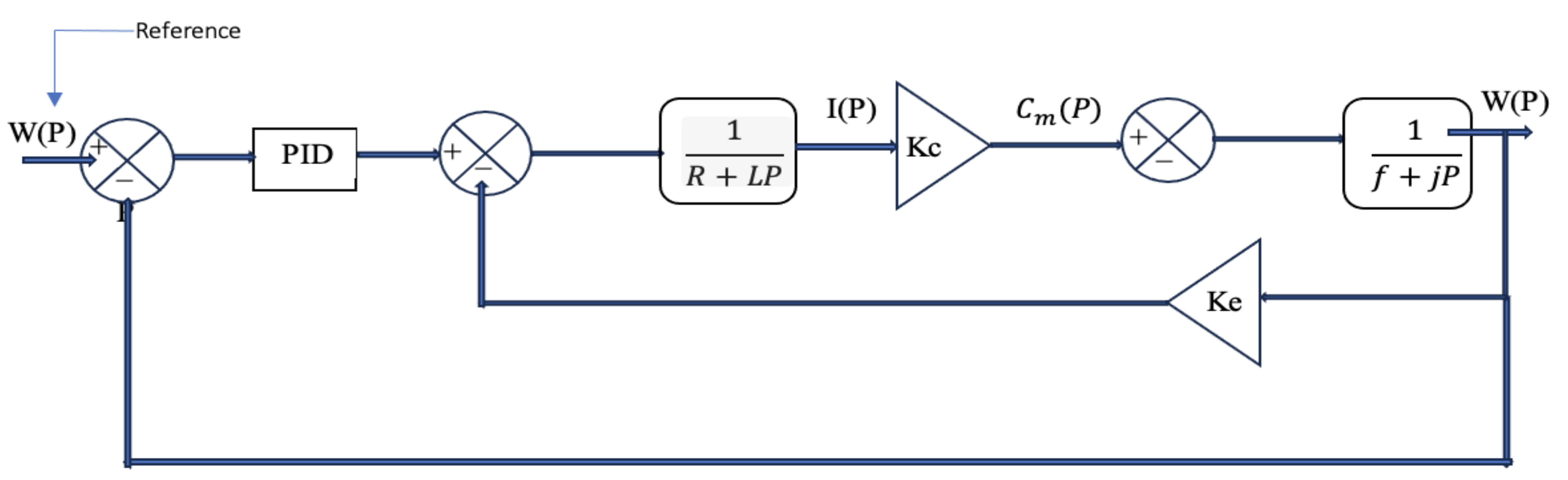
PID in Time Domain: $U(t) = K_P \times e(t) + K_i \times \int e(t)dt + K_d \times \frac{de(t)}{dt}$

$\Downarrow$

PID in Laplace Domain: $U(P) = K_P \times e(p) + \frac{K_i}{P} \times e(p) + P \times K_d \times e()$

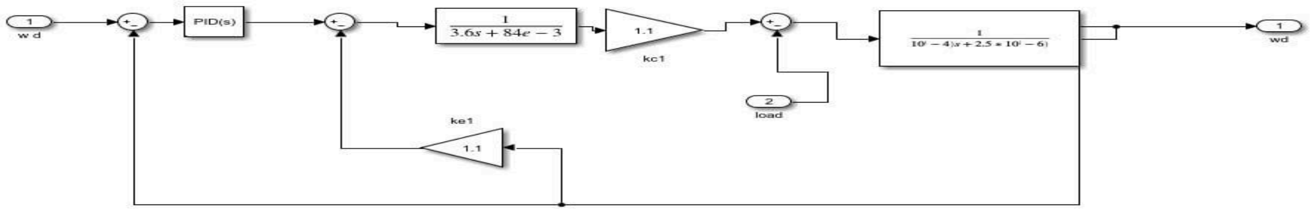
PID Speed Regulation of DC Motors

The behavior of the PID controller can be better understood through the following block diagram.
It illustrates how the controller processes the speed error to generate the control action.



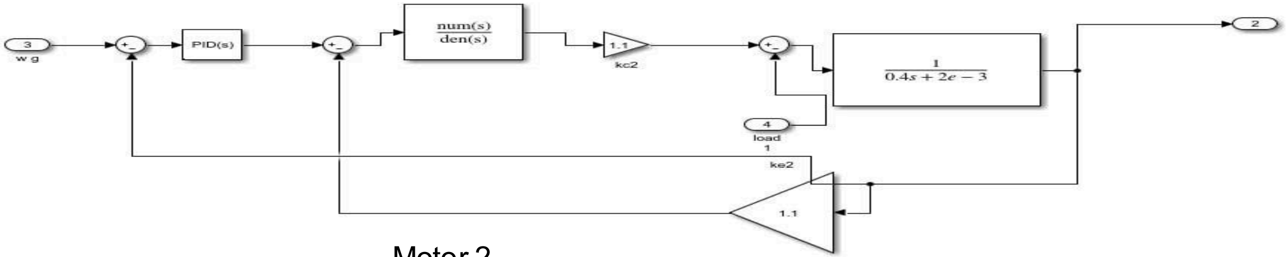
PID Speed Regulation of DC Motors

After introducing the PID control for one motor, we now apply it to both wheels of the robot. To control the full vehicle, we use two identical PID control blocks one for the left motor and one for the right motor.



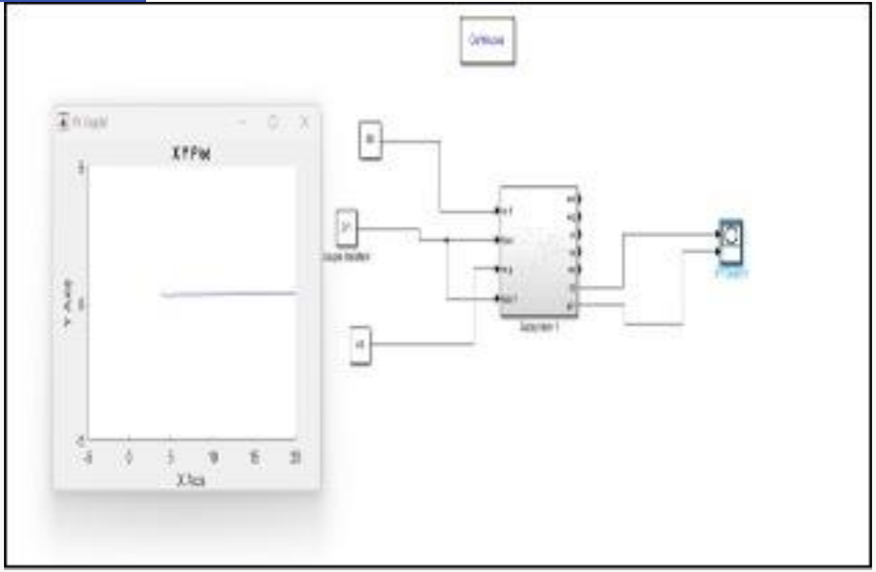
Motor 1

Each controller regulates the angular speed of its respective wheel.

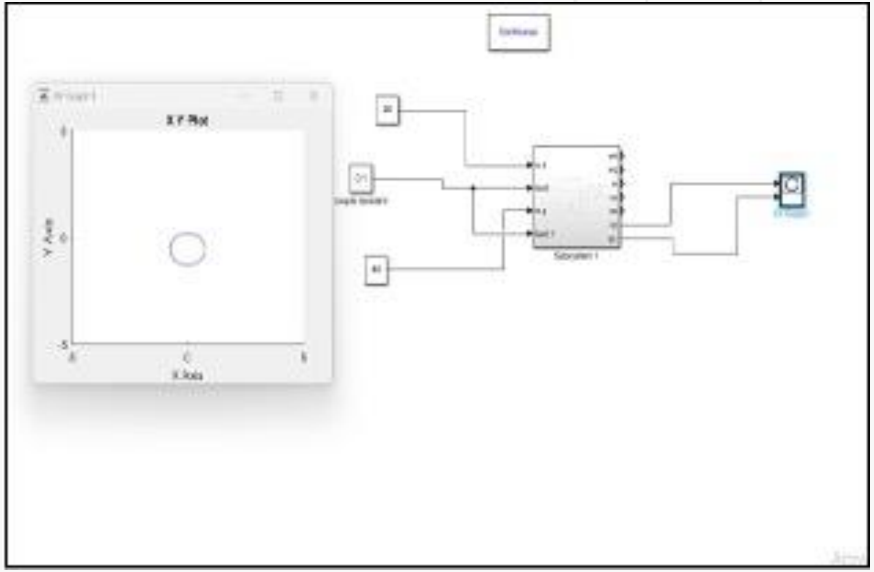


Motor 2

The following figures show two basic examples of such motion:



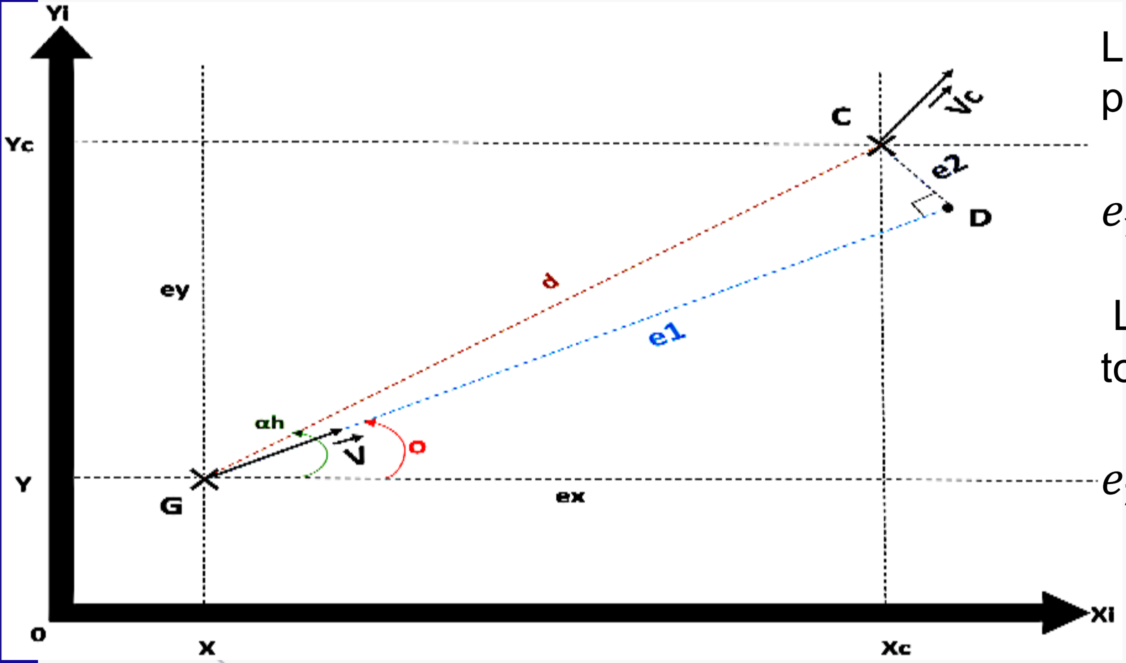
A straight path with ωr = ωl



A circular path with ωr ≠ ωl

Advanced Control Strategy: Following a Given Trajectory

In advanced control, the goal is to make the robot follow a specific path accurately. To do that, we calculate two types of errors that describe its position relative to the target trajectory.



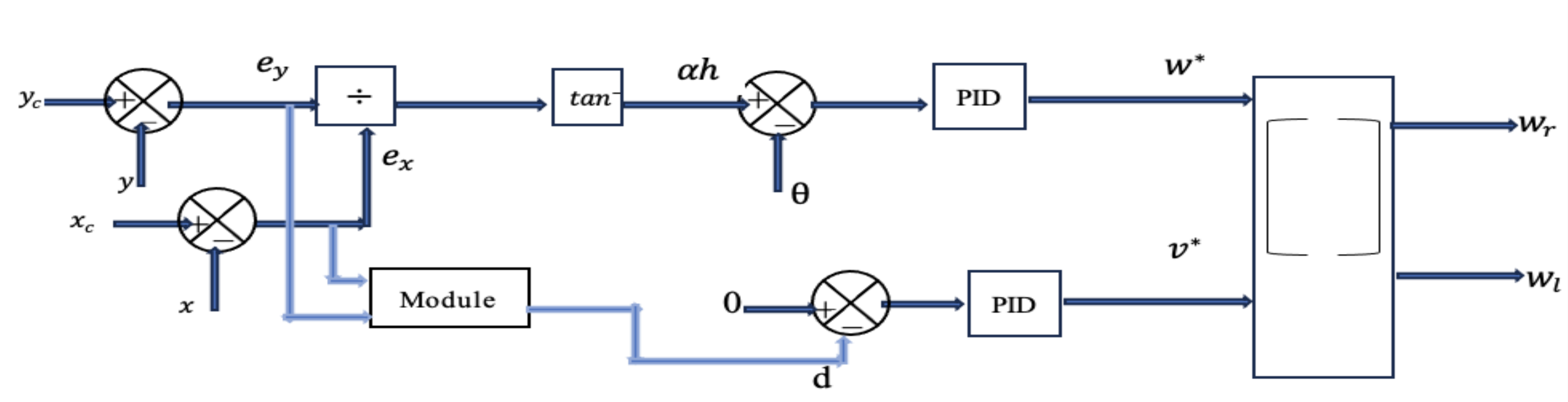
Longitudinal error (e_1): distance along the path

$$e_1 = d\cos(\alpha_h - \theta) = 0 \Rightarrow d = 0$$

Lateral error (e_2): distance perpendicular to the path

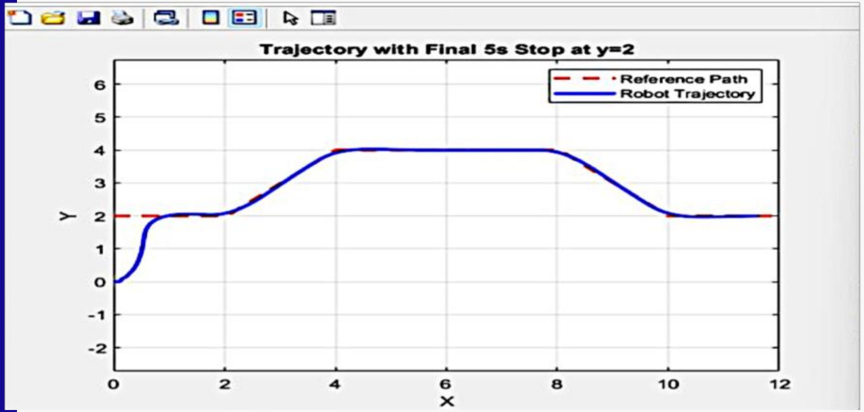
$$e_2 = d\sin(\alpha_h - \theta) = 0 \Rightarrow \theta = \alpha_h$$

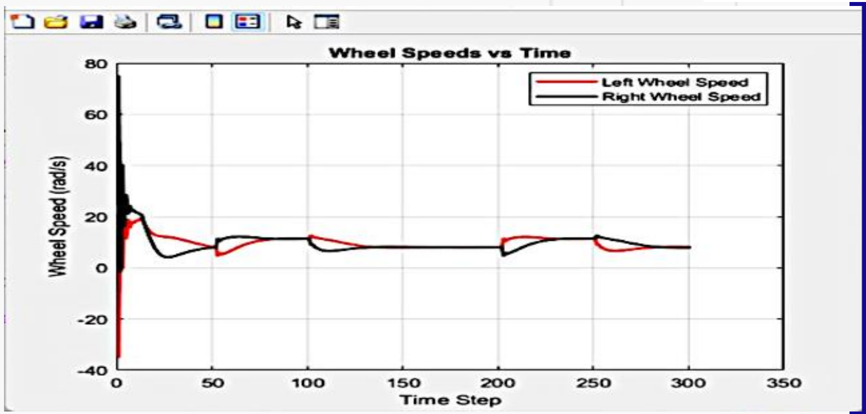
To ensure the robot accurately follows a desired trajectory, the control system must constantly evaluate its position relative to the reference path.
This is done by computing two errors: the longitudinal error (e_1) and the lateral error (e_2), which indicate how far the robot is from the path in both direction and alignment.



These errors are then used to generate appropriate linear and angular velocity commands (v and ω), allowing the robot to adjust its movement in real time and stay on track.

Simulation Results – Trajectory Tracking and Wheel Speeds



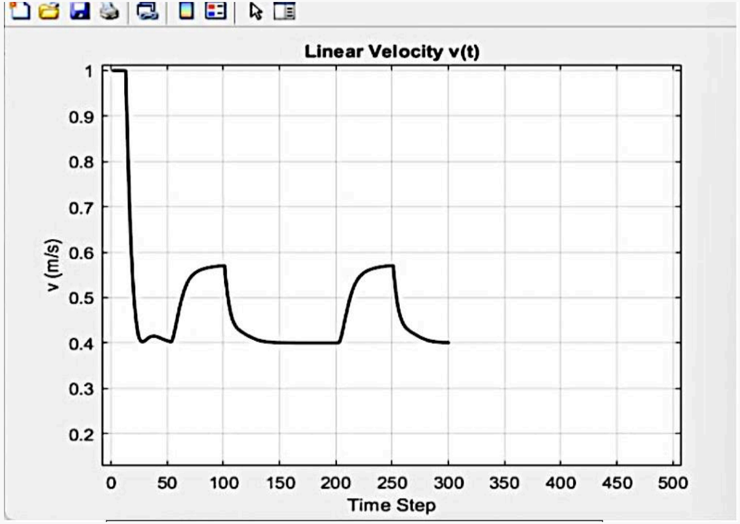


The first plot shows how the robot successfully follows the reference trajectory over time

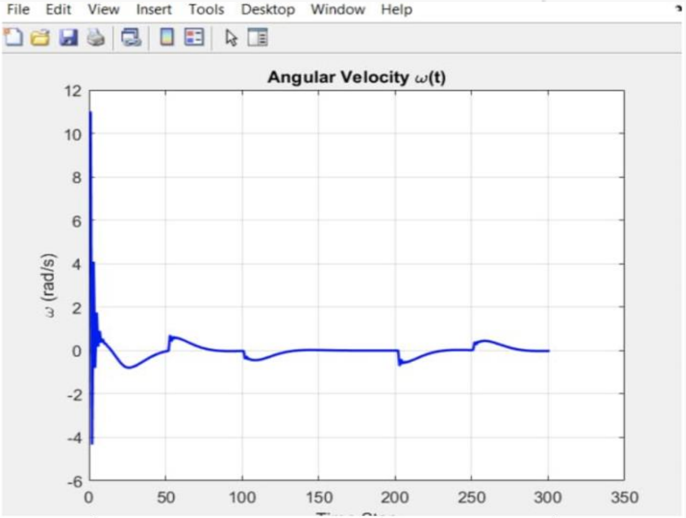
The second plot illustrates the evolution of each wheel's angular velocity (ωL and ωR), adapting constantly to keep the robot on track.

Simulation Results – Linear and Angular Velocities

These plots show how the robot adjusts its linear and angular speeds during trajectory tracking.



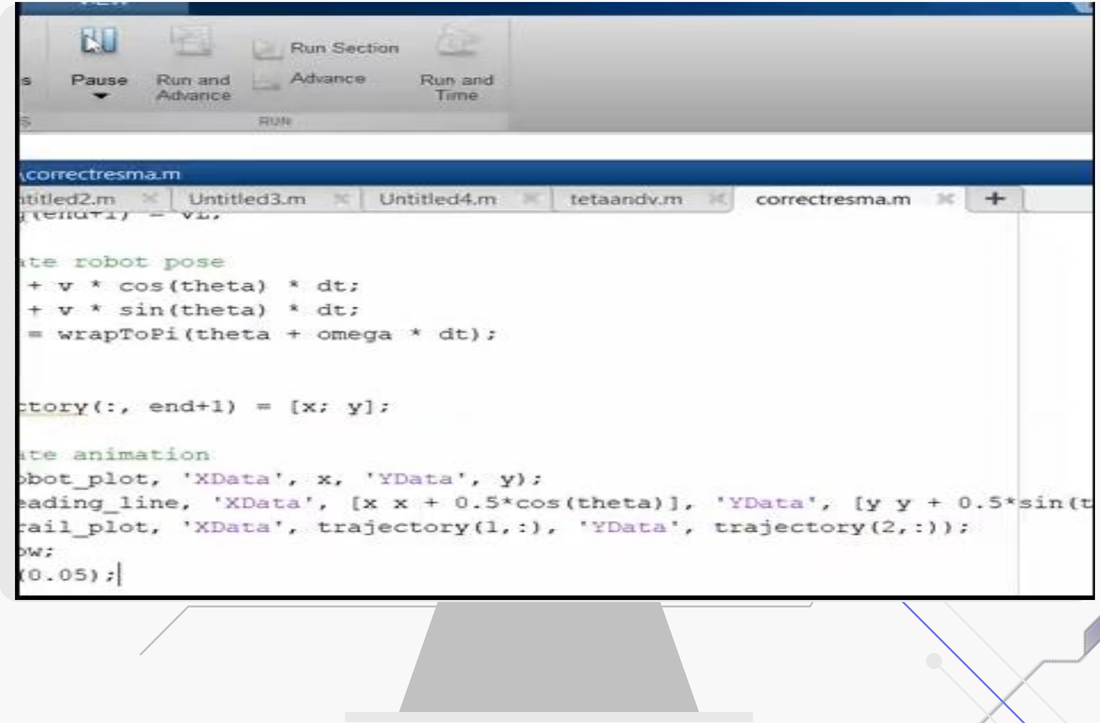
he linear velocity v(t) varies depending on whether the robot is moving straight or turning.



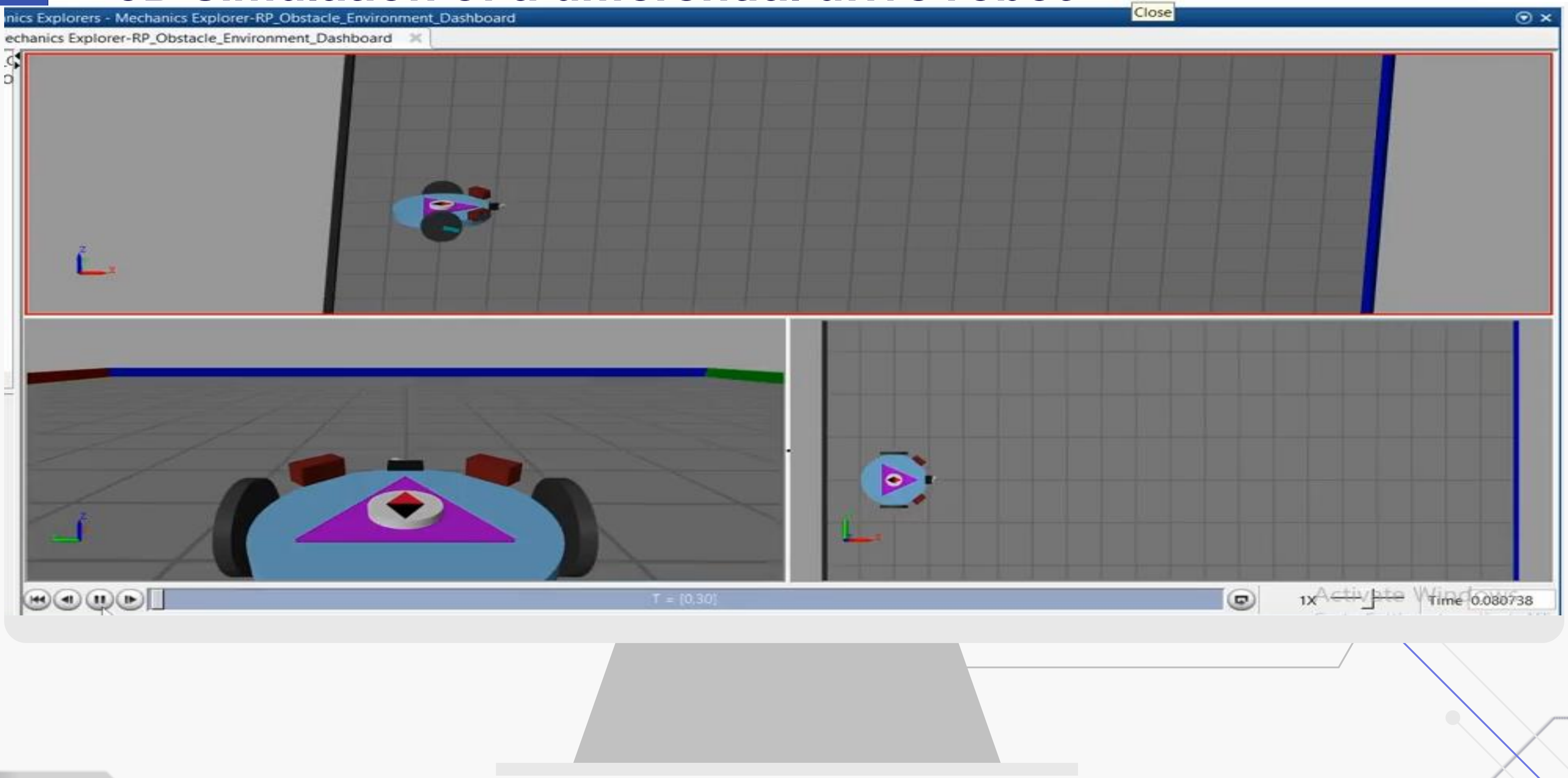
The angular velocity ω(t) increases when turning and drops near zero when the robot moves in a straight line.

Trajectory Tracking – Animation View

This animation shows how the robot follows the planned trajectory in real time.
It visually confirms that the robot remains aligned with the reference path using the implemented control strategy.



3D simulation of a differential drive robot



Conclusion

- ✓ We successfully modeled and simulated a differential-drive mobile robot.
- ✓ A PID-based control strategy was implemented to regulate wheel speeds and track a desired trajectory.
- ✓ Simulation results showed accurate path following and smooth dynamic behavior.
- ✓ The developed system can serve as a foundation for future real-world implementation or intelligent navigation techniques.

Thank You for Your Attention